When a Numbered Sketch Drops a Premise Used Twice

Working Title

When a Numbered Sketch Drops a Premise Used Twice.

Problem Statement

SketchThinker-R1's rewriting prompt requires an order-preserving numbered list, which has no mark for a visual fact used by two later steps. A linear list can treat that shared premise as a deletable detail, so the second use becomes a number with no justification.

Proposed Solution

This plan does not attach an existing reflection pruner. Graph-based chain-of-thought pruning removes low-impact reflections; it does not protect a premise whose out-degree is at least two, and it is text-only. Geometry items will be selected or rewritten so that one length or angle is a premise of two deductions, with that dependency annotated and valid alternative proofs accepted. A numbered-list sketch will be compared, at a matched length budget on Qwen2.5-VL-7B, with a sketch required to point again at any premise used twice. The measures are the drop rate of the shared fact and the accuracy of the second use. The proposition fails if those two quantities match.

Key Research Questions

- Are premises used twice dropped more often than premises used once?
- Does the accuracy loss sit on the second use rather than the first?
- Does requiring an explicit back-reference recover the second use inside the same length budget?

Expected Outcome

The study will show whether the numbered-list format, rather than brevity alone, deletes visual premises that later steps share.

Comparison

	Numbered list	Dependency sketch
Mark for a shared premise	None	A premise used twice must be pointed to again
What is counted	Drop rate of the shared fact; accuracy of the second use	The same two counts
Result that refutes it	—	Drop rate and second-use accuracy match the numbered list